Quantum Traffic Gridlock — traffic-derived QUBO and budgeted multi-angle QAOA

IBM Qiskit Fall Fest 2026 · QC@IITI. All costs below are computed by the official `scorer.py`. Code, outputs and logs are in the repository. Commands to reproduce every number are in `README.md`.

1. Traffic model and variables

There are 8 binary decisions, $x = (A_1 \ldots A_4, B_1 \ldots B_4)$, with $x_i = 1$ for NS green and 0 for EW green. One bit per intersection-slot encodes "exactly one phase is green", so **every bitstring is feasible**. No one-hot penalty, slack or auxiliary variable is needed, and the QUBO has 8 variables mapped to 8 qubits. The official cost decomposes exactly (verified against the scorer on 1,240 random pairs):

$C(x) = \Sigma_k D_k(x_{S_k}) + w_{sw} \Sigma_{n\in\{A,B\}} \Sigma_{t=2..4} (x_{n,t} + x_{n,t-1} - 2x_{n,t}x_{n,t-1})$,

where D_k = (delay + w_{ov}·overflow) of approach $k \in$ {A-NS, A-EW, B-NS, B-EW}. **Locality** fixes which bits each approach sees:

- the two A approaches see S_k = A bits;
- the unlinked B approach sees S_k = B bits;
- the linked B approach sees all 8 bits, because transfers round(f·departures) carry A's decisions into B one slot later.

Switching is already quadratic and is kept **exactly**.

2. QUBO derivation (scenario-independent modelling rule)

Why a surrogate is necessary. Each D_k is built from min(queue, capacity) (service) and max(0, queue − storage) (overflow). As a pseudo-Boolean function it has a unique multilinear expansion $D_k = \Sigma_S \hat{c}_S \Pi_{i\in S} z_i$ with $z_i = 1 - 2x_i$. Its degree is up to 4 within an intersection and up to 8 for the linked approach (spillback). Reducing all cubic-and-higher monomials would need far more than the 8 spare qubits. We therefore approximate each D_k by a quadratic function.

Rule. The QUBO is the weighted-L^2 projection of the exact traffic cost onto quadratic pseudo-Boolean functions:

$E = \text{argmin}_{g \text{ quadratic}} \Sigma_x \mu(x) (C(x) - g(x))^2$, with $g(x) = \text{offset} + \Sigma_{i\le j} Q_{ij} x_i x_j$.

- **Uniform μ (Walsh truncation).** This keeps exactly the Walsh coefficients with $|S| \le 2$, $\hat{c}_S = 2^{-8} \Sigma_x C(x) \chi_S(x)$ (orthogonality is asserted in the tests). The terms have a direct traffic meaning:
- $\hat{c}_{\{i\}}$ is the average marginal cost of giving NS rather than EW green in slot i;
- same-intersection $\hat{c}_{\{ij\}} > 0$ encodes diminishing returns (a green spent on an already-emptied queue is wasted, Fig. a);
- A–B couplings arise only from the linked approach, i.e. downstream spillback. Because projection is linear, the A-only approaches contribute no A–B terms.
- **Adaptive μ (used).** We set $\mu_\kappa(x) \propto \exp(-(E(x) - \min E)/(\kappa\cdot\sigma_E))$ and solve $E = \text{Proj}_{\{\mu(E)\}} C$ as a fixed point (8 iterations, $\kappa = 1$, frozen). The rationale is that QAOA puts its probability on low-energy states, so approximation error should be smallest there. The weights depend **only on the surrogate's own energies**. The true optimum is never computed or used; it is the scorer that judges the result. The weights stay broad: their effective sample size is 157 of 256 schedules on average (167 on R).

Computing the projection evaluates the traffic recurrence on its inputs: 16 local configurations for 4-bit approaches, 256 for the linked one. That is a modelling transform of the dynamics, not an optimisation. No schedule, coefficient or penalty is scenario-specific. The Ising form uses $x = (1 - z)/2$ and is stored in `qubo.json` with the variable↔qubit map.

Surrogate quality (QUBO exact minimizer evaluated by the official scorer):

	regret (QUBO min vs C*)	regret ≤ 5 %	QUBO min optimal	Pearson(E, C)
Scenario R	0 % (72.0 = C*)	—	yes	0.993
100 unseen scenarios, uniform μ	1.77 % mean, max 31 %	88 %	52 %	0.992
100 unseen scenarios, adaptive μ	**0.44 % mean, max 8.6 %**	**98 %**	**67 %**	0.989

On the rubric's 5 %→25 % scale, this is 9.67/10 (uniform) vs **9.97/10** (adaptive) expected surrogate-regret credit. Scenarios are drawn from the published hidden ranges (arrivals 0–8, capacity 4–6, storage 5–8, w_{ov} 2–6, w_{sw} 0.25–1.5, NS/EW link, f 0.4–0.8) and filtered to $C_{rand} - C^* > 5$.

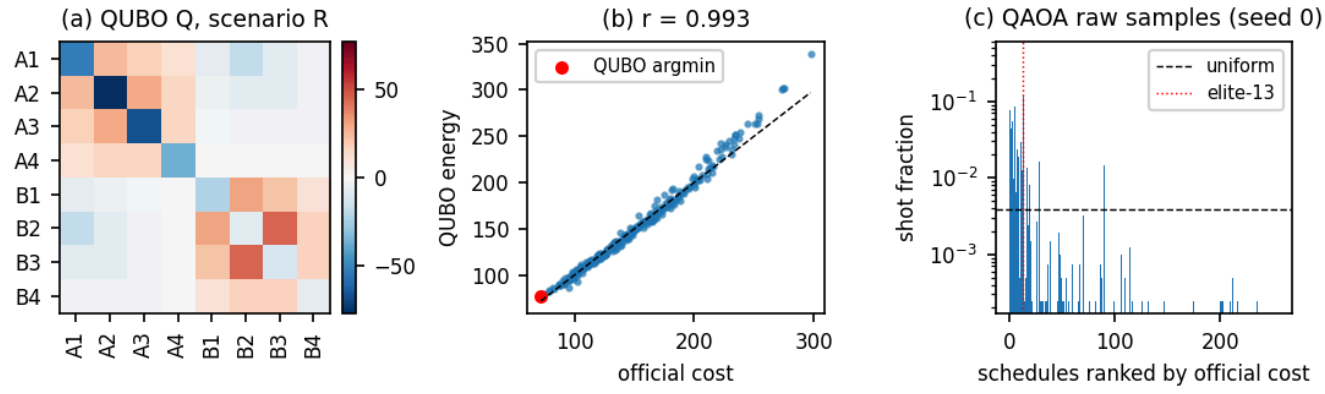


Figure 1. Scenario R: (a) QUBO matrix (symmetrised); (b) QUBO energy vs official cost for all 256 schedules; (c) 4,096 raw QAOA shots by official-cost rank (log scale).

3. QAOA design

Hamiltonian and normalisation. $H_C = \Sigma h_i Z_i + \Sigma J_{ij} Z_i Z_j$, divided by its spectral standard deviation σ (computed from h, J). This makes angles transfer across scenarios.

Ansatz (p = 4). $|\psi\rangle = \Pi_k U_M(\beta_k) U_C(\gamma_k) |+\rangle^{\otimes 8}$, with:

- $U_C = \exp(-i[\gamma_k^h \Sigma h_i Z_i + \gamma_k^J \Sigma J_{ij} Z_i Z_j])$: separate angles for fields and couplings;
- $U_M = \Pi_i \exp(+i \beta_{k,i} X_i)$: a **per-qubit (multi-angle) mixer**, with $H_M = -\Sigma X$ so that $|+\rangle$ is its ground state.

This gives 40 parameters. The circuit uses 8 qubits, 8 H, 32 RZ, 112 RZZ and 32 RX gates (depth 47). Standard QAOA is the special case of tied angles.

Initialisation. Three linear-ramp (Trotterised annealing) schedules with $\Delta t \in \{0.75, 1.0, 1.25\}$. Each costs one evaluation.

Objective. Gibbs objective $-\log\langle\exp(-\eta(E - E_{min})/\sigma_E)\rangle$ with $\eta = 2$ (Li et al. 2020). It rewards probability on low-energy states rather than only the mean. All objectives use **QUBO energies only, never the official cost.**

Optimiser and budget. COBYLA, two stages. Stage 1 optimises the standard 8-parameter ansatz up to evaluation 40. Its optimum is embedded into the 40-parameter ansatz (same state), which is refined to a hard cap of **300 evaluations**, probes included and asserted.

Training and sampling. Training uses an exact numpy statevector that implements the identical unitary. It agrees with Qiskit's `Statevector` of the exported circuit to ~1e-15, checked on every run. The final 4,096 shots come from Qiskit's `StatevectorSampler(seed)` on that circuit. Training is deterministic, so the seed only affects shot noise, which explains the tight seed spread. Runtime is about 2 s per seed.

Decoding. The raw-QAOA schedule is the sampled bitstring with the lowest QUBO energy (no scorer used). Separately labelled **hybrid**: steepest-descent 1-flip local search on the official cost, started from the 8 most frequent samples.

4. Results

Scenario R (C* = 72.0, C_rand = 151.25), 5 seeds (0–4):

method	official cost
(1) best alternating `10101010`	116.0
(2) greedy larger-queue `01111001`	92.5
(3) exact enumeration `11101010`	**72.0**
(4) uniform random expected	151.25
(5) QUBO exact minimizer (surrogate ceiling)	**72.0**
(6) QAOA raw samples: mean E_Q	81.3–82.1
(6) QAOA decoded schedule (every seed)	**72.0**
(7) HYBRID (labelled)	72.0

Over the 5 seeds, lift is 0.873 / 0.879 / 0.882 (min / mean / max) and enrichment is 15.14 / 15.37 / 15.56. The exact optimum receives 40–42 % of shots.

Generalization (100 unseen scenarios, generator seed 2026, frozen hyper-parameters):

- lift: mean 0.919, median 0.926, min 0.775;
- enrichment: mean 17.2, min 10.9 (random = 1, maximum 19.7);
- the exact optimum appears among the 4,096 samples in **100 %** of scenarios;
- fraction of the random→optimal gap closed: alternating 0.66, greedy 0.88, QAOA mean-of-samples 0.92, QAOA decoded schedule 0.994, hybrid 1.00.

Ablations (`results/ablations.md`; R + 40 unseen scenarios, generator seed 99, 4,096 shots):

variant (one change from final)	R lift	mean lift	min lift	mean enrich.	mean P(opt)
final (p=4, split γ, per-qubit β, Gibbs η=2, adaptive κ=1)	0.882	**0.909**	0.815	**16.85**	0.370
depth p = 1 / 2 / 3	0.727 / 0.907 / 0.885	0.717 / 0.886 / 0.907	0.543 / 0.778 / 0.817	10.7 / 16.1 / 16.7	0.14 / 0.38 / 0.37
standard ansatz (1 γ, 1 β per layer)	0.786	0.815	0.711	13.12	0.172
per-qubit β, single γ	0.910	0.884	0.744	16.24	0.327
objective ⟨E⟩ / $CVaR_{0.25}$	0.857 / 0.777	0.884 / 0.793	0.786 / 0.588	15.4 / 13.5	0.27 / 0.23
Gibbs η = 1 / 4	0.869 / 0.883	0.904 / 0.910	0.796 / 0.817	16.5 / 17.1	0.34 / 0.42
QUBO uniform μ (Walsh) / κ = 0.5 / κ = 2	0.847 / 0.883 / 0.876	0.899 / 0.913 / 0.905	0.789 / 0.819 / 0.790	16.1 / 17.1 / 16.7	0.36 / 0.37 / 0.37
init random angles / near-identity	0.867 / 0.828	0.857 / 0.889	0.654 / 0.765	14.9 / 15.9	0.28 / 0.33
budget 100 / 200 evaluations	0.778 / 0.847	0.809 / 0.872	0.689 / 0.773	12.6 / 15.4	0.16 / 0.27

Penalties: the encoding needs no constraint penalty (Section 1). The ablated weighting hyper-parameters are κ (surrogate measure) and η (objective temperature). The largest gains come from the multi-angle mixer (+0.07 lift), the annealing initialisation (+0.05 vs random) and the full 300-evaluation budget. η = 4 and κ = 0.5 look marginally better here but were **not** re-tuned on this reporting set; the frozen values stand.

5. Extension: noise analysis and noise-aware design (`experiments/noise.py`)

Noise model. The circuit is transpiled to {rz, sx, x, cx}. A 2-qubit depolarizing channel with rate e_2 follows each cx, a 1-qubit channel with rate $e_2/10$ follows each sx/x, and each readout bit flips with probability e_2. Probabilities are computed exactly with Aer density matrices on R plus 8 unseen scenarios.

Mean lift on R + 8 unseen scenarios (exact noisy distributions):

variant	cx	$e_2 = 0$	10^{-3}	$3{\cdot}10^{-3}$	10^{-2}	$2{\cdot}10^{-2}$	$5{\cdot}10^{-2}$
p = 1	56	0.716	0.705	0.683	0.613	0.523	0.319
p = 2	112	0.879	0.851	**0.799**	0.640	0.466	0.178
p = 4 (submitted)	224	**0.899**	0.847	0.752	0.495	0.270	0.042
p = 4, pruned τ = 0.1	115	0.881	**0.852**	0.797	0.635	0.461	0.182
p = 2, pruned τ = 0.25	35	0.797	0.788	0.771	**0.712**	**0.636**	**0.452**

Finding. The optimal depth is noise-dependent. p = 4 is best only when noiseless. At $e_2 \approx 3{\cdot}10^{-3}$ (current heavy-hex devices), the shallower p = 2 circuit wins, and pruning weak couplings ($|J| < \tau{\cdot}\max|J|$) trades a little noiseless lift for fewer CNOTs. On hardware we would therefore pick p and τ from a calibrated e_2 rather than fixing p = 4.

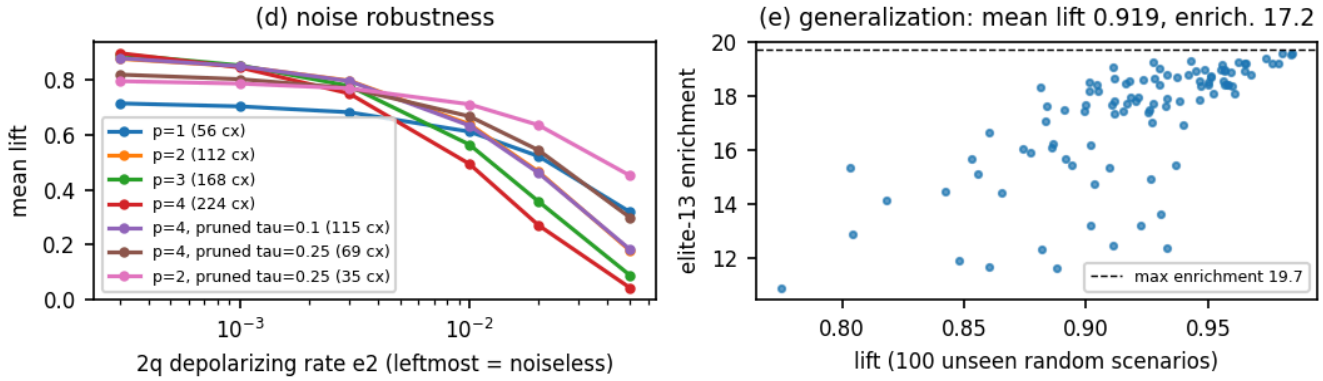


Figure 2. (d) mean lift vs depolarizing noise; (e) lift and enrichment of raw QAOA samples on 100 unseen scenarios.

6. Limitations

- The quadratic projection cannot represent the min/max hinges exactly. In 2 % of scenarios the QUBO minimizer is 5–9 % above C*, which bounds the decoded schedule (the hybrid repairs it). The lift ceiling of the surrogate itself is ≈ 0.99.
- Training uses exact expectations. With shot-based training, COBYLA would need a larger tolerance or SPSA. We expect lower lift at equal budget; this was not measured.
- The dense A–B coupling graph is the main noise cost (224 cx at p = 4). On a heavy-hex layout, SWAP routing would add more.
- The projection evaluates the recurrence on all 2^|S_k| local inputs. For long corridors the same rule applies per approach, but the adaptive weights would need Monte-Carlo estimation.
- Quantum advantage is neither claimed nor expected at 8 qubits. Exact enumeration of 256 schedules is trivial and is used only as a benchmark.